

Air-and-Water Heating System of DENERG Numerical Modelling



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Introduction

The aim of this project was to conduct a numerical analysis of the behavior of a fan coil unit. The configuration analyzed was derived from the typical layout and setup found in the offices of the Energy Department at Politecnico di Torino (DENERG). The primary objective was to simulate, using MATLAB and the numerical methods presented during the Numerical Modeling course, the behavior of the fan coil in terms of pressure variations, pressure gradients, temperature gradients, and temperature variations, under both transient and steady-state conditions, for both summer and winter operating scenarios.

These scenarios entailed different boundary and initial conditions. Given the complexity of accurately modeling a real fan coil unit, a simplification of the geometry was necessary to maintain computational feasibility while preserving the essential physical behavior. As such, the fan coil was modeled as a rectangular domain, focusing exclusively on the internal dynamics.

1. The Computational Domain

Initially, the fan coil was assumed to consist of ten vertical coils, equally spaced. However, during the modeling phase, it was determined that only three coils were sufficient to capture the essential dynamics, and this reduction proved beneficial in achieving a clearer and more manageable numerical output. Each coil was spaced 15.6 cm apart, had a width of 2 cm, and a height of 40 cm. A preliminary analysis was conducted to characterize the external casing material. Based on product catalogs, the casing was assumed to be made of PVC. Although this detail might appear secondary, it is in fact significant: the low thermal conductivity of PVC allowed us to assign a low α at the external boundaries, which is a critical parameter for heat exchange modeling.

While not explicitly represented in the simulation domain, the presence of a fan at the base of the fan coil was considered in the boundary condition setup. The fan plays a crucial role in the operation of the unit, as it drives air into the system, thereby increasing the inlet pressure. This effect was modeled indirectly, influencing both steady-state and transient pressure profiles. The outlet pressure was assumed to be approximately atmospheric, consistent with the need for conditioned air to be delivered into occupied spaces.

The computational domain was limited to half of the fan coil unit. The project was carried out by a team of eight students and was split into two subgroups. One group discretized the region between two coils using periodic boundary conditions, while our group exploited the symmetry of the system to analyze only the right half of the fan coil.

Finally, air properties such as thermal conductivity (λ), density (ρ), and specific heat capacity at constant pressure (C_p) were incorporated into the model. These param-

ters were essential for accurate modeling of the physical processes and for the correct implementation of boundary conditions.

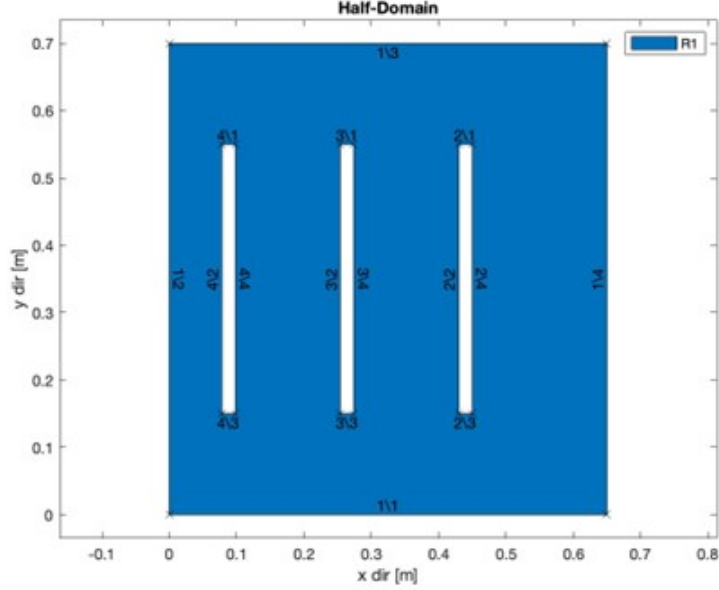


Figure 1: The domain representing half of the fan coil

2.1 Boundary Conditions for Pressure

Boundary conditions for the pressure field were defined identically for both winter and summer scenarios, since pressure variations between seasons were considered negligible in this study.

The assumptions and boundary conditions applied are as follows:

- An inlet flow enters the fan coil with a pressure of 10 atm. This relatively high value was deliberately chosen to produce clear and visible pressure gradients in the numerical simulations. Lower inlet pressures resulted in gradients that were not sufficiently distinct or visually appreciable.
- The outlet pressure was set equal to atmospheric pressure, providing a pressure drop across the fan coil unit.
- Along the left boundary, which represents the plane of symmetry of the system (and the interface with a mirrored computational domain), a Neumann boundary condition was imposed, enforcing zero normal pressure gradient:

$$\frac{\partial p}{\partial n} = 0.$$

- A similar Neumann boundary condition was applied along the right boundary of the domain, assuming that pressure gradients are significant only between the inlet and outlet regions:

$$\frac{\partial p}{\partial n} = 0.$$

Under these conditions, the pressure distribution within the domain exhibits a gradient predominantly along the flow direction (the Y-axis), driven by the pressure drop from the inlet to the outlet. This configuration reflects the influence of the fan, which increases the inlet pressure and maintains a directed airflow through the fan coil unit.

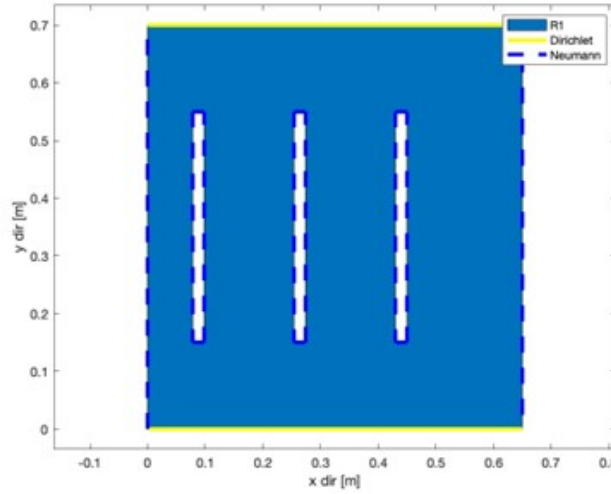


Figure 2: Boundary conditions regarding the Pressure evolution

2.2 Boundary Conditions for Temperature

Boundary conditions for temperature differed for winter and summer due to different thermal requirements.

Winter Scenario:

- External air temperature $T_{\infty} = 15^{\circ}\text{C}$.
Representing the temperature of the incoming air drawn from the outside environment. Although this value deviates from the one prescribed by official standards, it was deemed reasonable for the purposes of this simulation.
- Thermal conductivity of the medium $\lambda = 0.14 \text{ W}/(\text{m}\cdot\text{K})$.
- Convective heat transfer coefficient $\alpha = 0.1 \text{ W}/(\text{m}^2\cdot\text{K})$ (Due to the low thermal conductivity of PVC)
- $G = \alpha \cdot T_{\infty}$ was computed and used in the formulation of the boundary conditions.

The coils were assumed to heat the air from 15°C to 22°C, which corresponds to the target outlet temperature. Therefore, a **Dirichlet boundary** condition was imposed at the **inlet**, fixing the temperature to T_∞ . At the **outlet** and the **top boundary** of the domain, **Neumann boundary conditions** with zero thermal flux ($\frac{\partial T}{\partial n} = 0$) were applied. This assumes thermally insulated boundaries in those regions, consistent with the modeling approach, as directly prescribing the temperature at the outlet would not be meaningful in a transient or spatially varying simulation.

On the **left boundary**, representing the plane of symmetry, a **non-homogeneous Neumann** condition was applied. Since this boundary is shared with the mirrored half of the domain (simulated by the other group), the assumption of perfect symmetry implies that no net thermal flux crosses this boundary, effectively enforcing a continuity of temperature and thermal behavior across the plane.

On the **right boundary**, a **Robin boundary condition** was applied, as this side is exposed to the external environment. This condition accounts for heat exchange between the warmer internal fluid and the cooler ambient air outside the fan coil casing. The Robin condition incorporates both the ambient temperature T_∞ and the heat transfer coefficient α , allowing for a more realistic thermal interaction at the boundary with the PVC casing.

Summer Scenario:

For the Summer Conditions the same types of boundary conditions were used, but with different temperature values due to the cooling operation of the fan coil.

- Inlet air temperature $T_\infty = 32^\circ\text{C}$.
- Coil temperature fixed at 15°C for cooling.

The system exhibits a **negative temperature gradient** in contrast to winter's positive gradient.

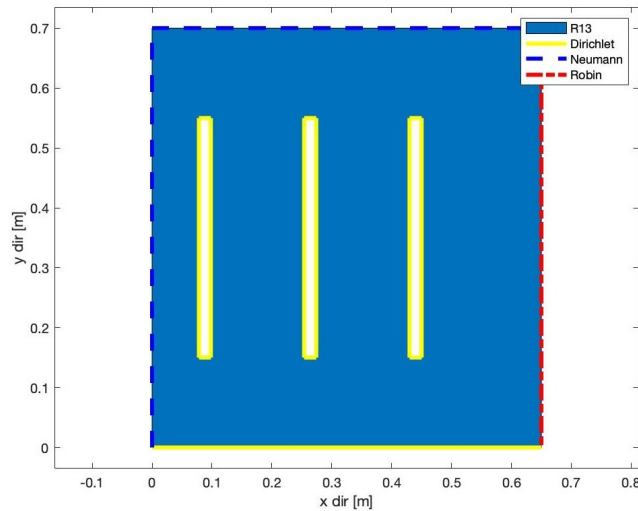


Figure 3: Boundary conditions for Temperature

3. Governing Equations

Heat Transfer Problem

Strong Form:

$$\rho c_p \frac{\partial T}{\partial t} - \nabla \cdot (\lambda \nabla T) = 0 \quad \text{in } \Omega$$

Boundary conditions:

- Dirichlet (inlet, coils): $T = T_{\text{boundary}}(t)$
- Neumann (symmetry, outlet): $-\lambda \frac{\partial T}{\partial n} = 0$
- Robin (PVC wall): $-\lambda \frac{\partial T}{\partial n} = \alpha(T_{\infty} - T)$

Weak Form:

$$\int_{\Omega} \rho c_p \frac{\partial T}{\partial t} v \, d\Omega + \int_{\Omega} \lambda \nabla T \cdot \nabla v \, d\Omega + \int_{\Gamma_R} \alpha T v \, d\Gamma = \int_{\Gamma_R} \alpha T_{\infty} v \, d\Gamma$$

Pressure Problem

Strong Form:

$$-\nabla \cdot (\nabla p) = 0 \quad \text{in } \Omega$$

Boundary conditions:

- Dirichlet: $p = p_{\text{boundary}}(t)$
- Neumann: $-\frac{\partial p}{\partial n} = 0$

Weak Form:

$$\int_{\Omega} \nabla p \cdot \nabla v \, d\Omega = 0 \quad \forall v \in V$$

These equations govern the heat and airflow behavior inside the fan coil unit modeled in this project. The heat transfer equation captures how air temperature changes due to heating or cooling from the coils, while the pressure equation ensures proper airflow driven by the fan. Boundary conditions reflect physical constraints, symmetry, and heat exchange with the environment. The equations were solved using finite element methods in MATLAB.

4. Matlab Results

Steady State

Pressure Field

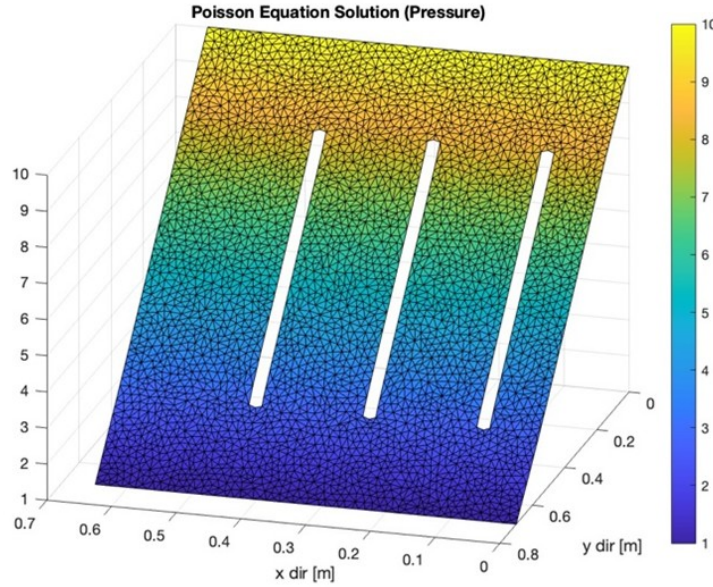


Figure 4: Pressure gradient visualization showing transition from inlet pressure to atmospheric pressure at the outlet.

As we can observe from the two images, the pressure problem is addressed firstly by visualizing the actual pressure gradient, which originates from the input pressure at the fan coil and decreases until reaching atmospheric pressure at the outlet. This outcome is confirmed both by the first image, which graphically illustrates the pressure variation through a color map — showing the transition from the initial pressure value to the atmospheric pressure at the outlet — and by the second image, which depicts the airflow using arrows to indicate the magnitude, direction, and orientation of the velocity vectors.

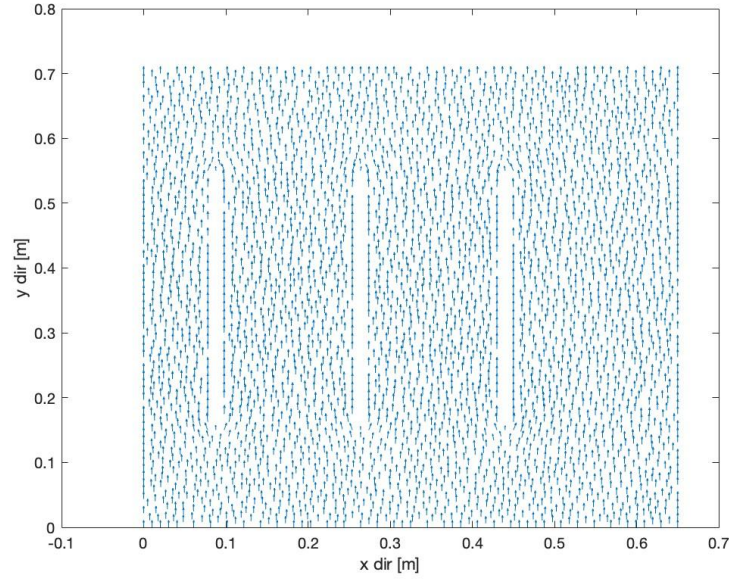


Figure 5: Vector field of airflow indicating magnitude, direction, and orientation of velocity vectors.

Given that homogeneous Neumann boundary conditions have been imposed along the lateral sides of the domain, there are no significant flows or pressure variations across the sides; instead, the primary variations occur along the Y direction. Consequently, due to the presence of the fan at the inlet of the fan coil unit, the air flow remains predominantly straight and parallel to the Y axis, resulting in a pressure drop from the imposed inlet value to atmospheric pressure at the outlet.

Temperature Profile – Winter

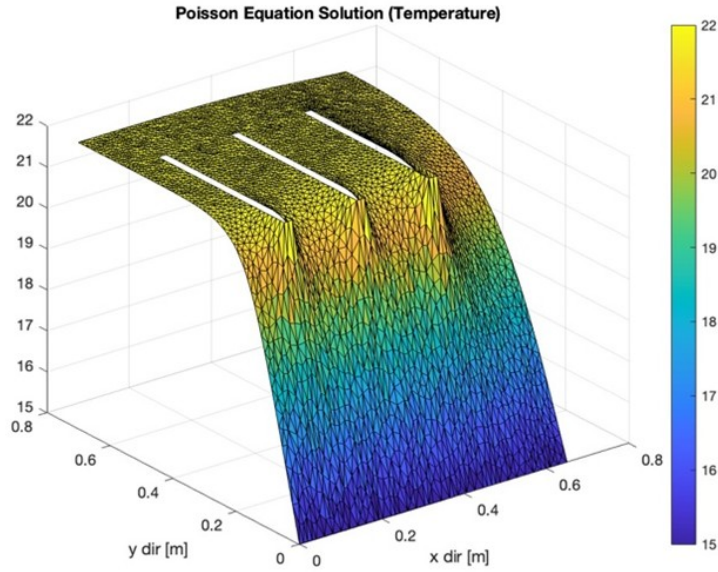


Figure 6: Temperature profile in steady state for winter conditions showing air heated from 15°C to 22°C.

Regarding the winter case, we can see from this image how the temperature problem has been successfully resolved, ultimately producing the desired output of 22°C.

It is evident that, under steady-state conditions, the air initially enters the domain at the ambient temperature T_∞ , which we have assumed to be 15°C. As the flow approaches the region containing the coils, which are preheated and maintained at a constant temperature of 22°C, the temperature progressively tends to equilibrate towards this higher value.

From the visualization, we can also observe that, along the right-hand side of the domain, despite the coils maintaining the temperature at 22°C, there is a slight drop in temperature, with values approaching approximately 21°C. This phenomenon is primarily due to heat loss, as some thermal energy is inevitably transferred from the air to the adjacent wall, which is in contact with the ambient air at $T_\infty = 15^\circ\text{C}$. Although PVC is a poor thermal conductor and thus a good insulator, a minor heat loss still occurs due to the airflow passing over the coils and moving toward the far-right side of the domain.

This slight temperature decrease along the right boundary is consistent with expectations and is accounted for in the simulation setup through the imposed boundary conditions, including the Robin boundary conditions that model convective heat exchange with the surroundings.

Temperature Profile – Summer

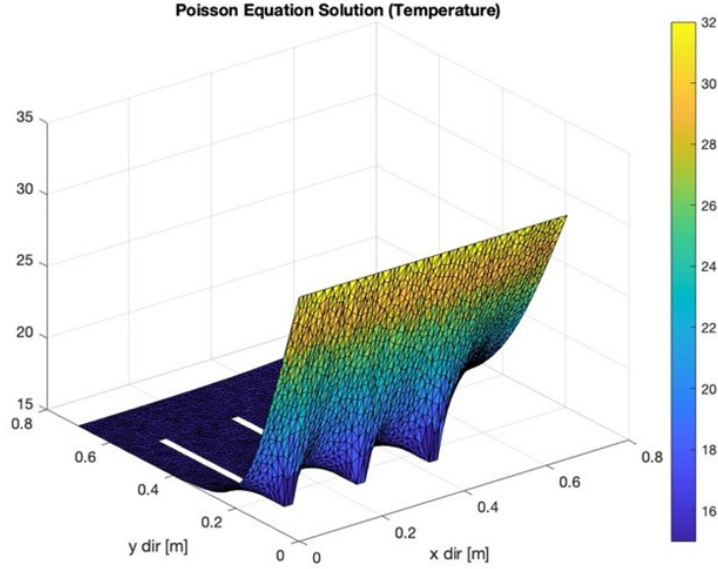


Figure 7: Temperature profile in steady state for summer conditions showing air cooled from 32°C to 15°C.

In the summer case, the process occurs in exactly the opposite manner. The temperature at the inlet is set to 32°C, as defined by our Dirichlet boundary conditions. As the air passes through the fan coil, driven by the fan, it is cooled by the coils, which are maintained at a constant temperature of 15°C. Consequently, the air temperature progressively decreases along the flow path.

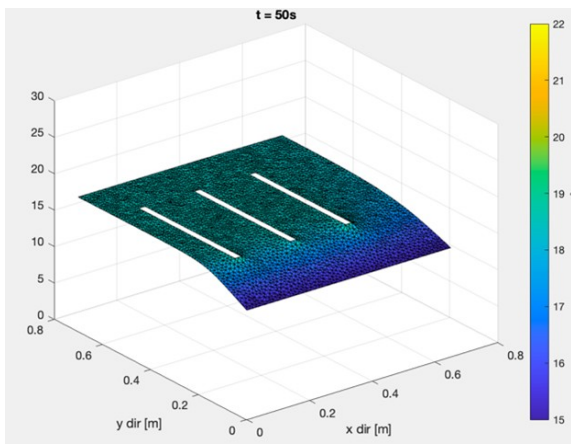
Under steady-state conditions, we observe that the temperature profile remains relatively uniform throughout the domain, except in the region in direct contact with the coils, where the cooling effect is most significant. As a result, the outlet temperature reaches the target value of 15°C, indicating effective heat exchange.

Regarding the boundary conditions, since they are identical to those implemented in the winter scenario, a minor temperature variation is observed near the right boundary. In this case, there is a slight increase in temperature due to heat gain from the surroundings. Although this effect is not clearly visible in the current image, it represents the minor thermal dispersion that occurs as the cooled fluid interacts with the adjacent wall, leading to a small heat gain despite the insulation of the system.

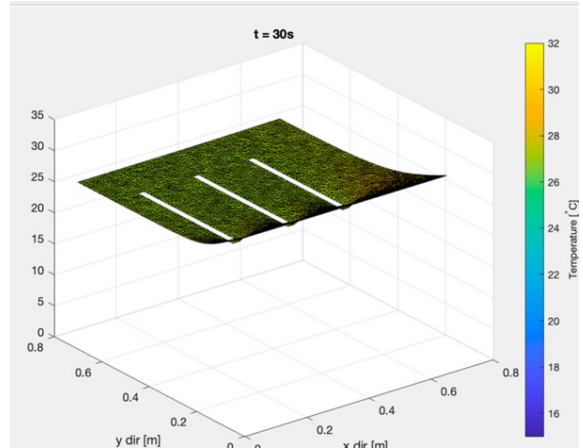
Transient Analysis

Regarding the transient analysis, the implemented modification consisted in considering the startup process for both scenarios, namely the summer and winter cases. In both situations, the initial condition assumes that the coils are at ambient temperature — 15°C for the winter case and 32°C for the summer case. A transient simulation was then conducted over a total duration of 300 seconds, within which the coils are brought to their steady-state operating temperature during the first 100 seconds.

Thus, the simulation can effectively be segmented into three distinct time points. These three specific time instants were selected arbitrarily; however, it is crucial to capture the transient behavior within the first 100 seconds. This initial transient phase illustrates how the air temperature inside the fan coil units evolves as the coil temperature itself increases or decreases toward its steady-state value.



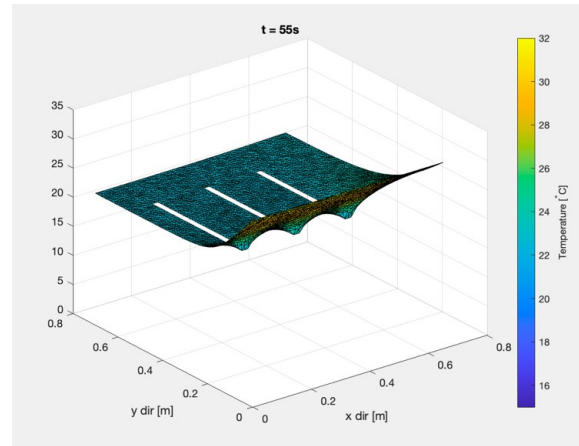
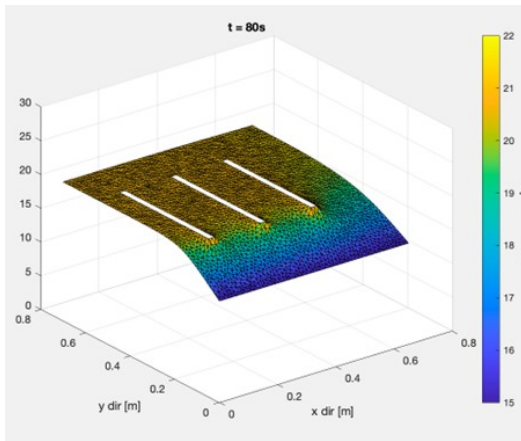
(a) Winter case at $t = 50\text{s}$.



(b) Summer case at $t = 30\text{s}$.

Figure 8: Temperature distribution in transient analysis for winter and summer scenarios.

The resulting simulations demonstrate that, as time progresses, the air temperature either rises or falls depending on the operating mode (heating or cooling). Consequently, we observe corresponding variations in the temperature profile within the domain, reflecting the thermal response of the system at each selected time instant.



These are the final results, equal to what is obtained by the steady-state simulations.

